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$$\sum_{n=0}^{\infty} \frac{2n+2}{\sqrt{n^3+5}}$$

6+ Verhoudigncriterium

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\frac{2n+2}{\sqrt{n^3+5}}}{\frac{1}{\sqrt{n}}} &= \lim_{n \rightarrow \infty} \frac{2n\sqrt{n} + 2\sqrt{n}}{\sqrt{n^3+5}} = \lim_{n \rightarrow \infty} \frac{(2n+2)\sqrt{n}}{\sqrt{n^3+5}} \quad (\sqrt{n^3+5} \sim n\sqrt{n}) \\ &= \lim_{n \rightarrow \infty} \frac{2n\sqrt{n}}{n\sqrt{n}} = 2 \Rightarrow 0 < 2 < \infty \\ &\Rightarrow \sum_{n=0}^{\infty} \frac{2n+2}{\sqrt{n^3+5}} \sim \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \\ \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} &= \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \text{ is een } p\text{-reeks met } p = \frac{1}{2} < 1 \Rightarrow \text{divergent, dus } \sum_{n=0}^{\infty} \frac{2n+2}{\sqrt{n^3+5}} \end{aligned}$$

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References